International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2019-0061
© 2019 Human Kinetics, Inc.

Human Kinetics
ORIGINAL INVESTIGATION

A Complex Relationship: Sleep, External Training Load, and Well-Being in Elite Australian Footballers

Benita J. Lalor, Shona L. Halson, Jacqueline Tran, Justin G. Kemp, and Stuart J. Cormack

Purpose: To assess relationships between objective sleep characteristics, external training loads, and subjective ratings of well-being in elite Australian football (AF) players. ***Methods:*** A total of 38 elite male AF players recorded objective sleep characteristics over a 15-day period using an activity monitor. External load was assessed during main field sessions, and ratings of well-being were provided each morning. Canonical correlation analysis was used to create canonical dimensions for each variable set (sleep, well-being, and external load). Relationships between dimensions representing sleep, external load, and well-being were quantified using Pearson *r*. ***Results:*** Canonical correlations were moderate between pretraining sleep and external training load ($r = .32–.49$), pretraining sleep and well-being ($r = .32$), and well-being and posttraining sleep ($r = .36$). Moderate to strong correlations were observed between dimensions representing external training load and posttraining sleep ($r = .31–.67$), and well-being and external training load ($r = .32–.67$). Player load and Player load 2D (PL2D) showed the greatest association to pretraining and posttraining objective sleep characteristics and well-being. Fragmented sleep was associated with players completing the following training with a higher PL2D. ***Conclusions:*** Maximum speed, player load, and PL2D were the common associations between objective sleep characteristics and well-being in AF players. Improving pretraining sleep quality and quantity may have a positive impact on AF players' well-being and movement strategy during field sessions. Following training sessions that have high maximum speed and PL2D, the increased requirement for sleep should be considered by ensuring that subsequent sessions do not start earlier than required.

Keywords: actigraphy, team sport, fatigue, player load

Australian football (AF) is a high-intensity, intermittent team sport involving tackling and collisions.[1] Monitoring training and competition loads is imperative to optimize performance[2] and can be conducted using session rating of perceived exertion (RPE) and microtechnology devices.[1] Self-report well-being questionnaires are a practical tool that can provide insight into an athlete's status in areas such as muscle soreness, stress, and injury.[3] The perceived well-being of AF players has been shown to influence the exercise intensity of subsequent field-based training sessions[4] and, therefore, may provide coaches an indication of the external load that can be expected during an AF training session.[2] A common component in these questionnaires is a subjective rating of sleep quality; however, recent work has demonstrated that this may have limited accuracy as an indicator of sleep quality in AF players.[5]

The prevalence of poor sleep in elite athletes is high,[6] although the objective sleep characteristics of AF players prior to and during competition have been shown to not be compromised compared with habitual sleep.[5] However, habitual sleep efficiency, an objective measure of sleep quality, was suboptimal (81.35%) in this population compared with recommendations for healthy adults (90%).[5] Although numerous factors including training and competition load may influence AF players' habitual sleep, the interaction between internal and external loads and the sleep characteristics of AF players has only been explored in a camp setting, where changes in session RPE showed a moderate negative correlation with sleep duration.[7] Importantly, the relationship between load and the sleep characteristics following the training session was not investigated. Higher training loads have been associated with small increases in sleep quality and quantity in other football codes during the preseason period.[8] However, this investigation did not include measures of subjective athlete well-being, and the commercially available sleep device utilized does not measure key characteristics of sleep, including sleep onset latency.

The aim of the study was to assess the relationships between objective sleep characteristics, external training load, and subjective ratings of well-being in elite AF players. Specifically, we aimed to (1) assess the association between sleep characteristics during sleep prior to a main training session and subsequent ratings of well-being and external load in the training, (2) determine if subjective ratings of well-being and measures of external load from training are associated with objective sleep characteristics in the subsequent sleep opportunity, and (3) assess if ratings of well-being are associated with the external loads during AF training.

Methods

Subjects

Data were obtained from 38 elite male AF players from the same team (mean [SD]; height 188.2 [7.3] cm; body mass 87.0 [8.1] kg; age 22.1 [3.3] y; AFL experience 60 [59] games played). Ethics approval was obtained from the Australian Catholic University human research ethics committee, conducted to conform to the recommendations of the Declaration of Helsinki.

Lalor, Halson, Kemp, and Cormack are with the School of Behavioural and Health Sciences, Australian Catholic University, Melbourne, VIC, Australia. Tran is with Knowledge Edge for Tokyo, High Performance Sport New Zealand, Auckland, New Zealand. Lalor (benita.lalor@myacu.edu.au) is corresponding author.

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC

Objective Sleep Characteristics

Participants undertook sleep monitoring over a consecutive 15-day period during the preparation phase of the season. Validated wrist-worn activity monitors were used to measure sleep (Philips Respironics Inc, Bend, OR).[9] Activity monitors were worn continuously (except during contact training and when in water), and data were recorded in 1-minute epochs. Participants recorded the start and end of each sleep period in a sleep diary.[10] The same sleeping environment was maintained throughout the monitoring period.

The combination of paper sleep diary and activity monitor data was used to determine when participants were awake or asleep. A detailed description of the scoring of sleep using the sleep diary and activity monitor has been previously provided.[5] The following information was collected from the activity monitors: bed time (hours:minutes), wake-up time (hours:minutes), sleep onset latency (minutes), sleep duration (hours), wake time (minutes), number of wake bouts, wake bouts average duration (minutes:seconds), and sleep efficiency (percentage). The sleep diary was used by participants to record sleep location, bed time (hours:minutes), and wake-up time (hours:minutes). The sleep diary was reviewed at regular intervals during the monitoring period.

External Training Load

A total of 14 training sessions occurred during the 15-day period. This included 5 main field, 3 conditioning, 2 cross-training (boxing and cycling), and 4 strength training sessions. External training load was assessed for main field sessions only, all starting at 9 AM and followed by the same hydrotherapy and nutrition recovery protocols. Main field sessions consisted of a standardized warm-up, 2 closed skill drills, 2 full ground skill drills, followed by 3 match simulation drills. No players were modified from drills during the monitoring period.

During the field-based training sessions, players wore the same microtechnology device (Minimax version 4; Catapult Innovations, Melbourne, VIC, Australia), sampling at 10 Hz.[1,11] The device sat firmly between the shoulder blades in a vest worn underneath the training jumper. Each device also housed an internal triaxial accelerometer sampling at 100 Hz. The summation of the triaxial accelerations is determined as Player Load™ (PL).[1] PL and player load 2D (PL2D), an accumulation of data from the anterioposterior and mediolateral axes only, and player load slow, a metric representing activity (eg, grappling) performed at velocities < 2 m/s, were also recorded. These variables are demonstrated to have acceptable levels of reliability. Catapult sprint (version 5.1.0.3; Catapult Innovations, Melbourne, Australia) software was used to download data. External training load metrics were expressed in absolute form and relative to training duration, as outlined in Table 1.

Self-Reported Measures of Well-Being

Self-reported measures of soreness (calf, quad/knee, groin/hip, and lower back), mood, sleep quality, sleep duration, leg heaviness, and energy levels were collected each morning prior to any training. Well-being questions were completed (95% compliance) using a Likert scale ranging from 1 (poor) to 10 (good) via an application (Smartabase; Fusion Sport, Queensland, Australia) on players' phones.[12]

Statistical Analysis

Exploratory data analysis confirmed that most variables met parametric assumptions. Visual inspection of quantile–quantile plots showed that those variables that did not meet parametric assumptions presented with increasing residuals as the values of the variable increased. Square root transformations can be effective for variables with such a data structure, so variables that violated parametric assumptions were transformed and then retested. The following variables did not meet parametric assumptions even after transformation and thus were excluded from the primary analysis: activity monitor measured bed time, player load slow (per minute), high-intensity running (meters), and high-intensity running (meters per minute). Descriptive statistics (median ± interquartile range) were calculated for all sleep, well-being, and external load variables.

Figure 1 outlines the associations tested in the canonical correlation analysis (CCA). The variables used in each "variable set" are presented in Table 1.

Canonical correlation analysis is a multivariate approach that quantifies the relationships between 2 sets of variables and is used to quantify relationships between multiple variables simultaneously.[13] This multivariate technique was selected because each of the primary measures in this study comprises multiple variables that, in combination, represent a person's sleep, external load, or well-being. To quantify relationships between 2 variable sets, the CCA procedure produces functions that are akin to multiple linear regressions.[13] Each function uses the variables from each variable set as inputs to 2

Table 1 Variable Sets Used in Canonical Correlation Analysis

External training load	Objective sleep characteristics	Self-reported measures of well-being
Absolute loads	Sleep onset latency, min	Mood, AU
Total distance, m	Sleep duration, h:min	Calf soreness, AU
Player load, AU	Sleep efficiency, %	Leg heaviness, AU
Player load 2D, AU	Wake time, min	Quad/knee soreness, AU
Player load slow, AU	Wake bouts, n	Groin/hip soreness, AU
Maximum speed, km/h	Wake bouts average duration, min:s	Lower back soreness, AU
HIR distance, m	Wake-up time, h:min	Sleep duration, AU
Loads relative to training duration		Energy levels, AU
Average velocity, m/min		Hamstring soreness, AU
Player load/min, AU		Sleep quality, AU
Player load 2D/min, AU		
HIR distance, m/min		

Abbreviations: AU, arbitrary unit; HIR, high-intensity running >24 km/h.

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC

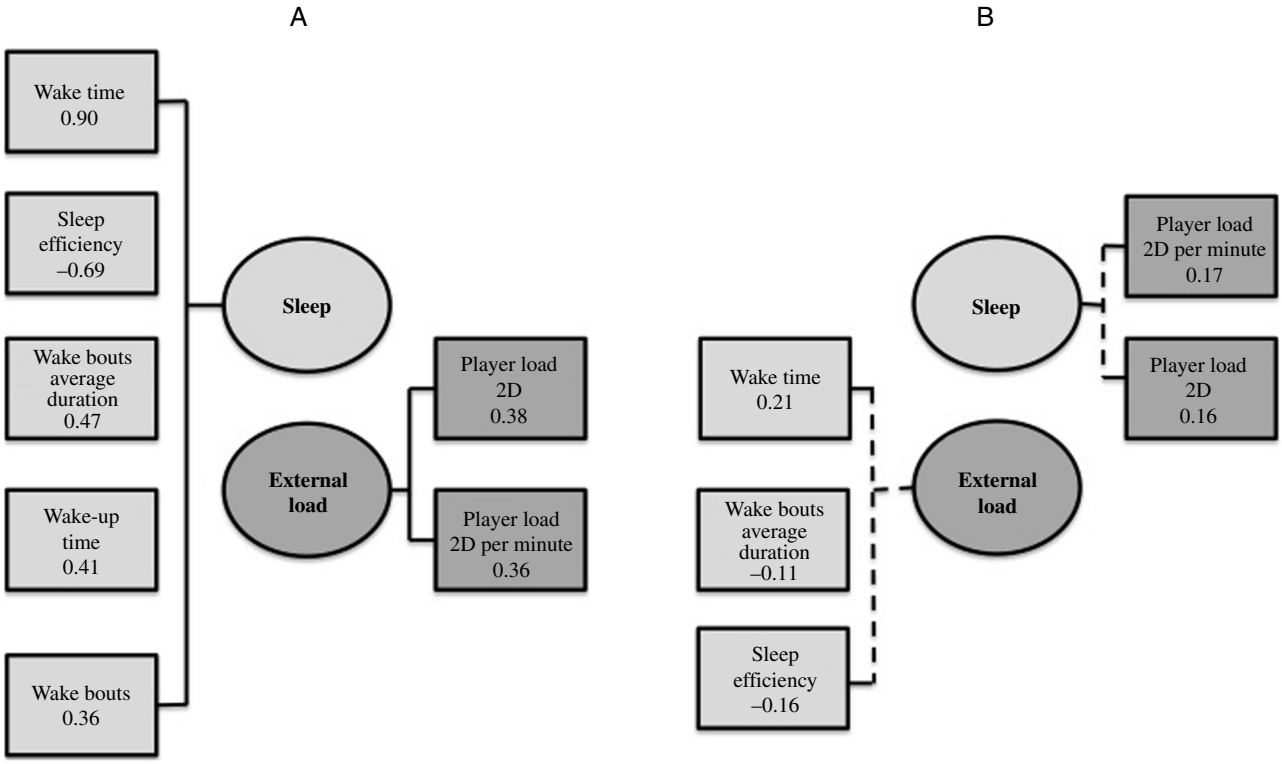

Figure 1 — Associations tested in the canonical correlation analysis. Canonical correlations were used to examine the associations between (A) pretraining sleep and subsequent pretraining well-being, (B) pretraining sleep and subsequent external training load, (C) pretraining well-being and subsequent external training load, (D) well-being and posttraining sleep, and (E) external training load and posttraining sleep.

regressions (one for each variable set). Each regression produces 1 synthetic variable (ie, 1 canonical dimension, or simply, dimension) for each variable set. The aim of these functions is to maximize the correlation strength between any pair of dimensions. In addition, this analysis will produce as many dimensions as there are variables in the smaller variable set.[13] The dimensions are distinct from each other as they arise from different linear combinations of the same input variables and must meet the condition of double orthogonality, where "both synthetic variables in subsequent canonical functions must be uncorrelated with both synthetic variables in all preceding functions."[13] Readers seeking more details about this procedure and its application are encouraged to review the primer on CCA written by Sherry and Henson.[13]

For example, consider a CCA of the sleep and well-being variables. The first canonical function will produce sleep dimension 1 and well-being dimension 1. Sleep dimension 1 is a linear combination of multiple sleep variables; the same concept applies for the well-being dimension 1. The weighting or contribution of each variable to its dimension is adjusted in the analysis to maximize the correlation strength between sleep dimension 1 and well-being dimension 1. The second canonical function will produce sleep dimension 2 and well-being dimension 2. In this function, the analysis again adjusts the weighting of each input variable to satisfy the condition of double orthogonality. Because of this condition, the strongest possible canonical correlation will always present between the dimension 1 pairing, whereas subsequent canonical correlations express relationships given the residual variance in the variable sets that were unexplained by the preceding canonical function(s).

Key results produced by a CCA include: (1) tests of dimensionality, (2) canonical loadings (also known as structure coefficients), and (3) cross-loadings. Tests of dimensionality produce the canonical dimensions, the canonical correlation coefficients between dimensions, and the statistical significance of each canonical correlation. The strength of relationships between dimensions (ie, canonical correlations) is expressed using Pearson correlation coefficient (r), interpreted in accordance with established thresholds[14] were: weak: $-.3 > r \leq -.1$ and $.1 \geq r < .30$; moderate: $-.5 > r \leq -.3$ and $.3 \geq r < .50$; and strong: $r \leq -.5$ and $r \geq .5$. Canonical correlations that are statistically significant ($P < .05$) are presented in the results. Canonical loadings are also expressed as Pearson r, where the coefficient represents the relationship between each variable in a variable set and the canonical dimension for the same variable set. The loading of each variable onto its canonical dimension is calculated in the presence of all other contributing variables to the same canonical dimension; as a result, canonical loadings are not affected by multicollinearity.[15] Cross-loadings represent the relationship between each variable in 1 variable set and the canonical dimension for the other variable set.[13]

All analyses were conducted using R (version 3.5.1)[16] and the following packages: "dplyr" and "reshape2" for data preparation,[17,18] "Multivariate normality (MVN)" for assumption testing,[19] and "CCA" for the canonical correlation analyses.[20]

Results

Sleep Characteristics, External Training Load, and Well-Being

Sleep characteristics, external load from 5 on-field training sessions, and self-reported measures of well-being are presented in Table 2.

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC

Table 2 Objective Sleep Characteristics, External Training Load, and Self-Reported Measures of Well-Being

Parameter	Median ± IQR
Sleep characteristics	
Sleep onset latency, min	5 ± 19
Sleep duration, h:min	8:32 ± 1.29
Sleep efficiency, %	81.90 ± 9.14
Wake time, min	77 ± 40
Wake bouts, n	31 ± 11
Wake bouts average duration, min:s	02.30 ± 0.39
Bed time, h:min	22:43 ± 1.17
Wake-up time, h:min	07:23 ± 1.14
External load field training sessions	
Absolute loads	
Total distance, m	5997 ± 2572
Player load, AU	593 ± 304
Player load 2D, AU	355 ± 175
Player load slow, AU	174 ± 100
Maximum speed, km/h	27.9 ± 3.2
HIR distance, m > 24/km/h	266 ± 342
Loads relative to training duration	
Player load, AU/min	10.31 ± 5.80
Player load 2D, AU/min	6.31 ± 3.67
Player load slow, AU/min	2.98 ± 1.22
Average velocity, m/min	111 ± 58
HIR distance, m/min > 24/km/h	3.33 ± 5.93
Self-reported measures of well-being, AU	
Mood	7 ± 1
Calf soreness	7 ± 2
Leg heaviness	7 ± 1
Quad/knee soreness	7 ± 1
Groin/hip soreness	7 ± 1
Lower back soreness	7 ± 2
Sleep duration	8 ± 1
Energy levels	7 ± 2
Hamstring soreness	7 ± 1
Sleep quality	8 ± 1

Abbreviations: AU, arbitrary unit; HIR, high-intensity running; IQR, interquartile range.

Pretraining Sleep and External Training Load

The relationships between pretraining sleep and external training load metrics are represented by 3 canonical correlations. The strongest canonical correlation was between sleep dimension 1 and external training load dimension 1 ($r = .49$, $P = 4.26 \times 10^{-28}$). Figure 2A shows the variables with the highest loadings onto each dimension; wake time was the stronger contributor to the sleep dimension 1 ($r = .90$), whereas PL2D and PL2D per minute contributed similarly to external training load dimension 1 ($r = .38$ and .36, respectively).

Figure 2B presents the cross-loadings of each variable onto dimension 1 for the other variable set.

Pretraining sleep and external training load dimensions 2 and 3 are presented in Table 3.

Pretraining Sleep and Well-Being

The relationship between pretraining sleep and well-being is represented by 1 canonical dimension ($r = .32$, $P = .002$). Figure 3A shows the variables with the highest loadings onto the dimension; wake-up time was the strongest contributor to the sleep dimension 1 ($r = .63$), whereas self-reported sleep quantity contributed most to the well-being dimension 1 ($r = .60$).

Figure 3B presents the cross-loadings for this dimension.

External Training Load and Posttraining Sleep

The relationship between external training load and posttraining sleep is represented by 4 canonical correlations. The strongest canonical correlation was between external training dimension 1 and sleep dimension 1 ($r = .67$, $P = 5.06 \times 10^{-43}$). Figure 4A shows the variables with the highest loadings onto each dimension; wake-up time was the strongest contributor to sleep dimension 1 ($r = -.55$), and total distance was the strongest contributor to external training load dimension 1 ($r = -.82$). The cross-loadings for dimension 1 are presented in Figure 4B. Total distance ($r = -.46$) showed a moderate negative correlation with sleep dimension 1. Dimensions 2, 3, and 4 are presented in Table 4.

Well-Being and Posttraining Sleep

The relationship between pretraining well-being ratings and posttraining sleep is represented by 1 canonical correlation ($r = .36$, $P = .002$).

Figure 5A presents the variables with the highest loadings onto dimension 1; wake-up time was the strongest contributor to sleep dimension 1 ($r = .61$), whereas self-reported sleep quality contributed most to well-being dimension 1 ($r = -.61$). Figure 5B presents the cross-loadings of each variable into dimension 1 for the other variable set.

Well-Being and External Training Load

The relationship between pretraining well-being and subsequent external training load measures is represented by 4 canonical correlations. The strongest canonical correlation was between well-being dimension 1 and external training load dimension 1 ($r = .67$, $P = 3.6 \times 10^{-58}$). Figure 6A shows the variables with the highest loadings onto each dimension; hamstring soreness was the strongest contributor to well-being dimension 1 ($r = -.55$), and PL was the strongest contributor to external training load dimension 1 ($r = -.74$). Figure 6B presents the cross-loadings of each variable into dimension 1 for the other variable set. PL ($r = -.48$) showed a moderate negative correlation with well-being dimension 1. Dimensions 2, 3, and 4 are presented in Table 5.

Discussion

This study revealed that complex relationships exist between sleep, external training load, and well-being measures in AF players. Fragmented sleep the night prior to training was associated with lower mood, increased soreness, and players completing the following training session with higher PL2D. Furthermore, players who completed training with a lower total distance were associated with a later wake-up time in the next sleep opportunity. Our findings highlight that the relationship between objective sleep characteristics, external training load, and well-being is not

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC

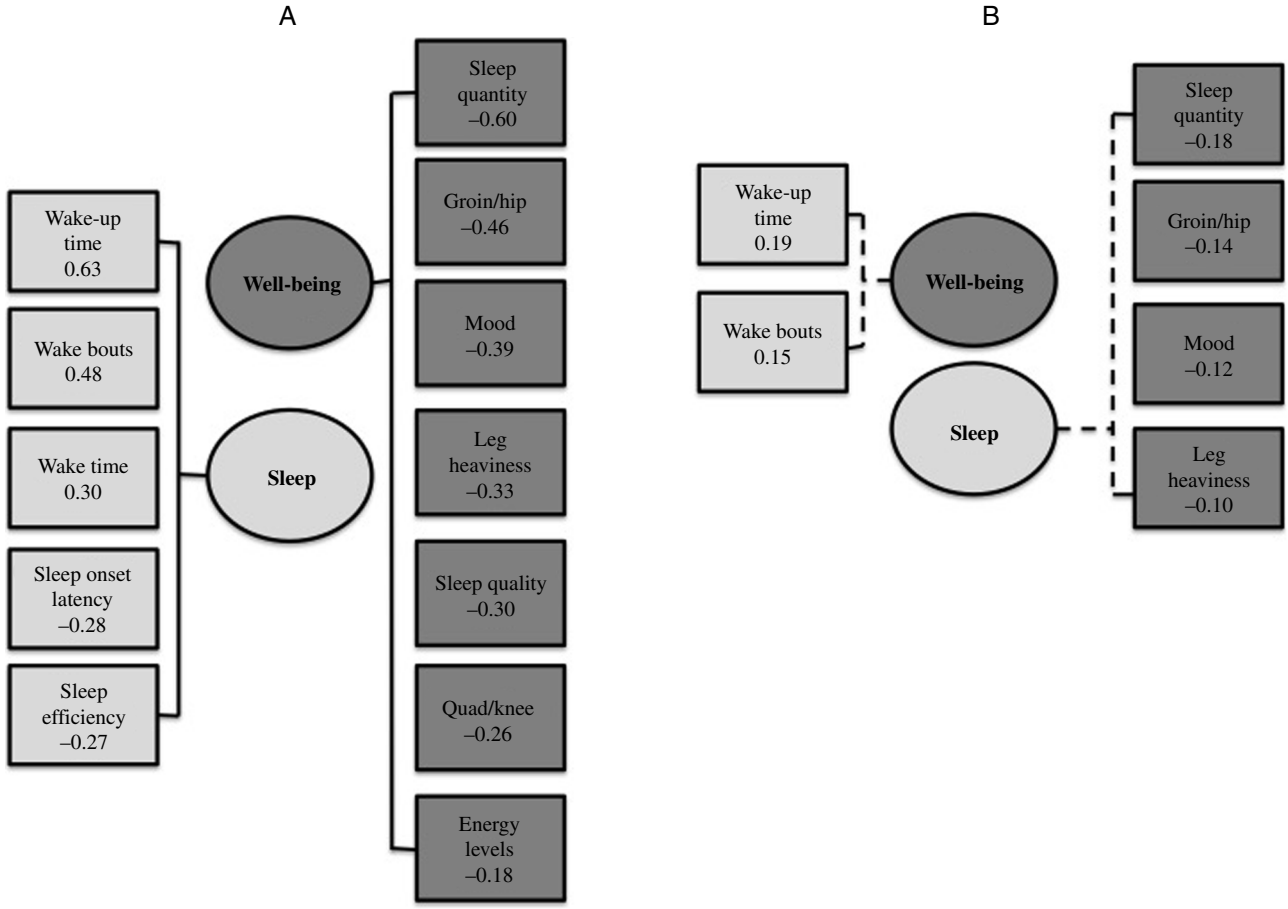

Figure 2 — (A) Pretraining sleep and external training load dimension 1* ($r = .49$, $P = 4.26 \times 10^{-28}$) and (B) pretraining sleep and external training load 1 cross-loadings. *The dimension is statistically significant ($P < .05$).

Table 3 Pretraining Sleep and External Training Load Canonical Correlations

	Dimension 2[a] ($r = .46$, $P = 2.99 \times 10^{-15}$)		Dimension 3[a] ($r = .32$, $P = 6.91 \times 10^{-5}$)	
	Canonical loading	Cross-loading	Canonical loading	Cross-loading
Sleep characteristics				
Sleep onset latency, min	–.81[b]	–.18	–.23	
Sleep duration, min			.54	
Sleep efficiency, %				
Wake time, min	.15			
Wake bouts, n		-		
Wake bouts average duration, min:s	.21		.27	
Wake-up time, h:min	–.40		.68[b]	.11
External load variables				
Absolute loads				
Total distance, m	.13		–.48[b]	–.18
Player load, AU	.30	.14	–.13	
Player load 2D, AU	.29	.13		
Player load slow, AU			.16	
Maximum speed, km/h	.59[b]	.27		
Loads relative to training duration				
Average velocity, m/min	.12			
Player load/min, AU	.57	.27	.39	.15
Player load 2D/min, AU	.56	.26	.47	.18

Abbreviation: AU, arbitrary unit. Note: The relationship between external training load metrics and pretraining sleep characteristics. Data are expressed as canonical correlations with the canonical loading and cross-loading of each variable presented for each dimension.
[a]The dimension is statistically significant. [b]The variables from each data set that had the strongest contribution to each dimension.

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC

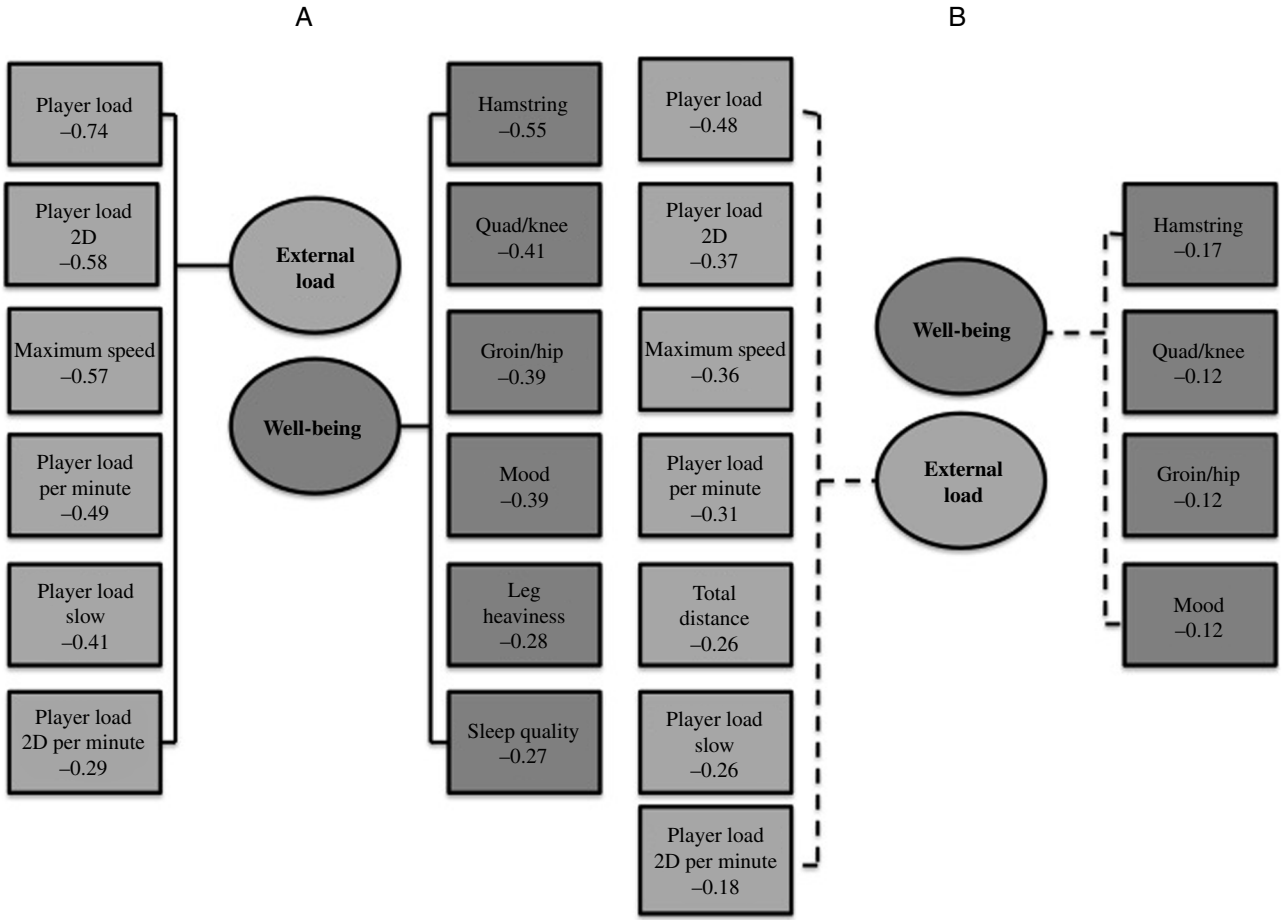

Figure 3 — (A) Pretraining sleep and well-being dimension 1 ($r = .32$, $P = .002$) and (B) pretraining sleep and well-being dimension 1 cross-loadings.

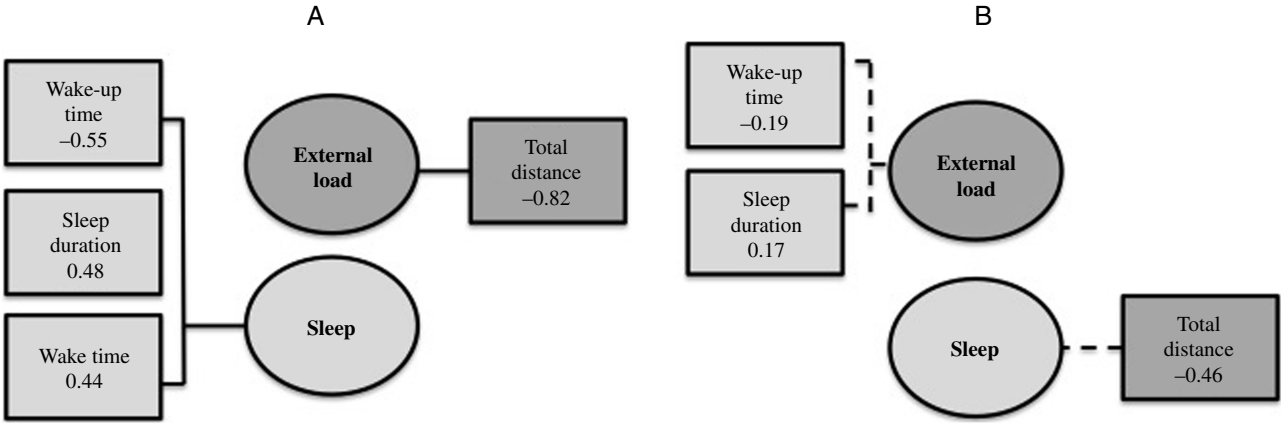

Figure 4 — (A) External training load and posttraining sleep dimension 1* ($r = .67$, $P = 5.06 \times 10^{-43}$) and (B) external training load and posttraining sleep and dimension 1 cross-loadings. *The dimension is statistically significant ($P < .05$).

defined by 1 measure (eg, sleep duration). Instead, the contribution of a number of sleep variables including wake time, number of wake bouts, and wake-up time may influence both external load and the well-being of AF players. These findings have implications for practitioners, particularly when choosing variables to monitor AF players' sleep to assist in the planning and evaluation of training.

The strongest canonical dimensions for pretraining sleep and external training load, indicated AF players with a lower sleep efficiency and a longer wake time in the pretraining sleep, had a higher PL2D during training. These associations may be due to players entering the training session tired from fragmented sleep or from an accumulated sleep debt, resulting in players performing high-intensity movements to compensate for inefficient running or

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC

Table 4 External Training Load and Posttraining Sleep Canonical Correlations

	Dimension 2[a] ($r = .50$, $P = 1.85 \times 10^{-17}$)		Dimension 3[a] ($r = .43$, $P = 4.27 \times 10^{-8}$)		Dimension 4[a] ($r = .31$, $P = 1.28 \times 10^{-2}$)	
	Canonical loading	Cross-loading	Canonical loading	Cross-loading	Canonical loading	Cross-loading
Sleep characteristics						
Sleep onset latency, min	–.80[b]	–.21	–.42		–.12	
Sleep duration, min			–.21		.64[b]	.10
Sleep efficiency, %	.55	.14	–.43	–.10	.49	
Wake time, min	–.49	–.13	.65[b]	.14		
Wake bouts, n	–.29		.10		–.11	
Wake bouts average duration, min:s	–.11		.57	.13		
Wake-up time, h:min			.27		.60	
External load variables						
Absolute loads						
Total distance, m			.24	.10	.16	
Player load, AU	.14				.14	
Player load 2D, AU			.24	.10	.21	
Player load slow, AU			–.15			
Maximum speed, km/h	.43	.22	.60[b]	.25	–.45[b]	–.14
Loads relative to training duration						
Average velocity, m/min					–.39	–.13
Player load/min, AU	.54[b]	.27			.33	.11
Player load 2D/min, AU	.38	.19	.29	.12	.40	.13
HIR distance, m/min > 24/km/h	.29	.15				

Abbreviations: AU, arbitrary unit; HIR, high-intensity running. Note: The relationship between external training load metrics and posttraining sleep characteristics. Data are expressed as canonical correlations with the canonical loading and cross-loading of each variable presented for each dimension.
[a]The dimension is statistically significant. [b]The variables from each data set that had the strongest contribution to each dimension.

poor positioning.[2] In support of this, AF players are shown to have an altered activity profile in a fatigued state, resulting in players becoming inefficient in the production of load (greater lateral or vertical movements).[2] In addition to changes in movement strategy when tired, extended wake periods have been associated with effects on brain function (including attention lapses), cognitive function, and performance.[21] There is potential for players with a low sleep efficiency, due to extended wake time, to experience impaired cognitive function during the training session.[21] Even relatively moderate sleep restriction, if sustained night after night, can impact neurobehavioral functioning.[22] In contrast, extending total sleep time can improve reaction time, daytime sleepiness, and mood in collegiate athletes.[23] As a result, improving sleep quality and quantity may have a positive impact on AF players' performance, particularly when psychomotor vigilance is required.[23] Objective monitoring of sleep characteristics of AF players may also provide an indication of the external load that might be expected in the subsequent training session and provide practitioners with the ability to make adjustments to training drill participation if warranted. Practitioners should specifically monitor PL and PL2D in players who have reported poor sleep and potentially modify training participation.

The canonical correlation between pretraining sleep dimension 1 and well-being dimension 1 highlighted the potential impact of fragmented pretraining sleep on subjective well-being. An increased number of wake bouts in the sleep opportunity prior to the main training session was associated with lower mood and increased soreness. Similar findings have been observed in elite cyclists during an intensive training period when cyclists who experienced an increased number of wake bouts reported disruptions to mood state, increased tension, fatigue, confusion, depression, and stress.[24] An increase in total sleep time has also been associated with an improved self-perception of performance during training and competition.[23] This supports the idea that obtaining extra, good quality sleep is likely to have beneficial effect on an athletes' overall well-being and mental readiness to train.[23] Practitioners may improve well-being and reduce perceptions of fatigue during the preseason training period by optimizing both the quality and quantity of AF player's sleep. This may be achieved through minimizing sleep disturbances and wake time by prioritizing sleep hygiene strategies.[25]

The second aim of this study was to examine the associations between well-being, external training load, and posttraining sleep characteristics. The canonical correlation between well-being dimension 1 and posttraining sleep dimension 1 indicated that higher ratings of soreness and shorter subjective sleep quantity were associated with fragmented sleep in the posttraining sleep opportunity. This was due to longer wake duration and longer wake bouts. There is limited information regarding the interactions between external training load, muscle soreness, and sleep characteristics in elite sport, and this warrants further investigation. To assist athletes in improving the quality of sleep following a training session, practitioners should consider strategies that may improve the recovery from sessions involving contact (eg, cold-water therapy, use of compression garments, massage, and nutrition strategies).[26] The relationship between

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC

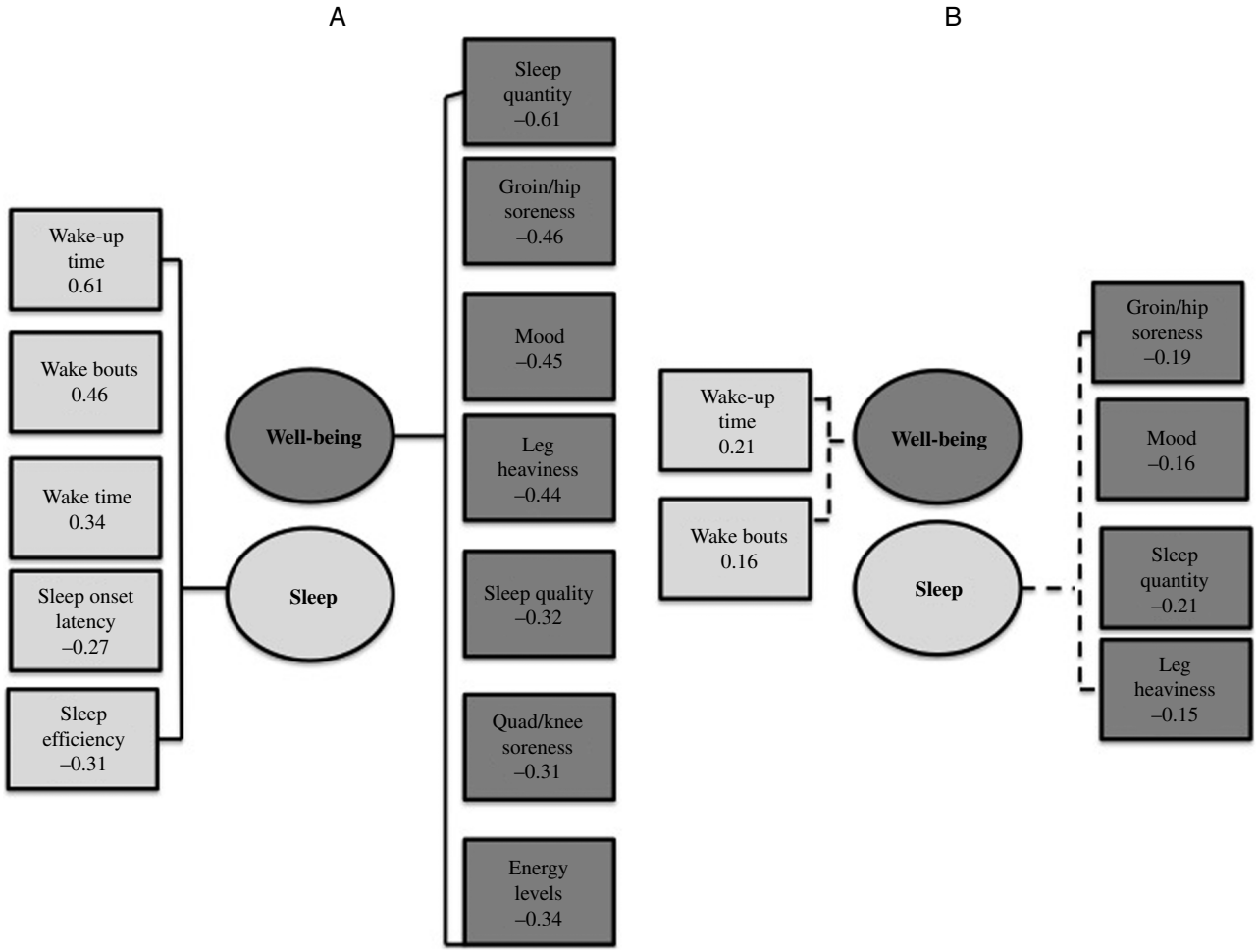

Figure 5 — (A) Well-being and posttraining sleep and dimension 1* ($r = .36$, $P = .002$) and (B) well-being and posttraining sleep 1 cross-loadings. *The dimension is statistically significant ($P < .05$).

external training load dimensions 1 and 2 and sleep dimensions 1 and 2 highlighted that the content of the training session may affect the requirement for sleep following the session in AF players. First, lower total distance in training was associated with a later wake-up time in the sleep opportunity following the training session. Our findings also indicated that a higher maximum speed and higher PL per minute in training were associated with a shorter sleep onset latency and higher sleep efficiency in the posttraining sleep. A similar relationship was observed in youth soccer players, where an increased sleep duration and later time of final awakening were observed in the sleep opportunity following a training session involving an increased high-speed distance.

It is possible that training that produces higher maximum speed increases an AF player's requirement for sleep. However, it is acknowledged that these associations could also be due to between player variations in both training design factors, such as role within a training drill or positional group requirements and also interindividual differences in sleep characteristics. Further research is required to understand the associations between external training load in intermittent team sports and objective sleep characteristics. Practitioners should maximize the opportunity for sleep following sessions that have high speed by ensuring that sessions the next day do not start earlier than required.

The final aim of this study was to assess the relationship between pretraining perceived well-being and the external load produced in the main training session. Previous work has demonstrated the influence of pretraining well-being scores on the external load (specifically PL and player load slow) during AF skill-based training sessions.[4] Our results suggest that players who reported an increased lower body soreness (hamstring, quad/knee, and groin/hip) and lower rating of mood prior to the main training session were associated with a higher PL, PL2D, and a higher maximum speed when compared with other players in the team. This supports the suggestion that pretraining well-being (specifically lower body soreness and low mood) may also be associated with an inefficient production of load (greater lateral or vertical movements or increased sprinting to compensate poor decision making) during the session. For an AF player, mood alteration also has the potential to affect motivation to train and willingness to give maximal effort in both training and competition.[27] It has also been demonstrated that perception of effort increases under sleep deprivation, which can influence an athlete's time to fatigue during training and competition.[27] Collection of these measures prior to AF training may provide practitioners an indication of the external load that can be expected during the session and may guide strategies to improve players' mood and subsequent training and competition output.

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC

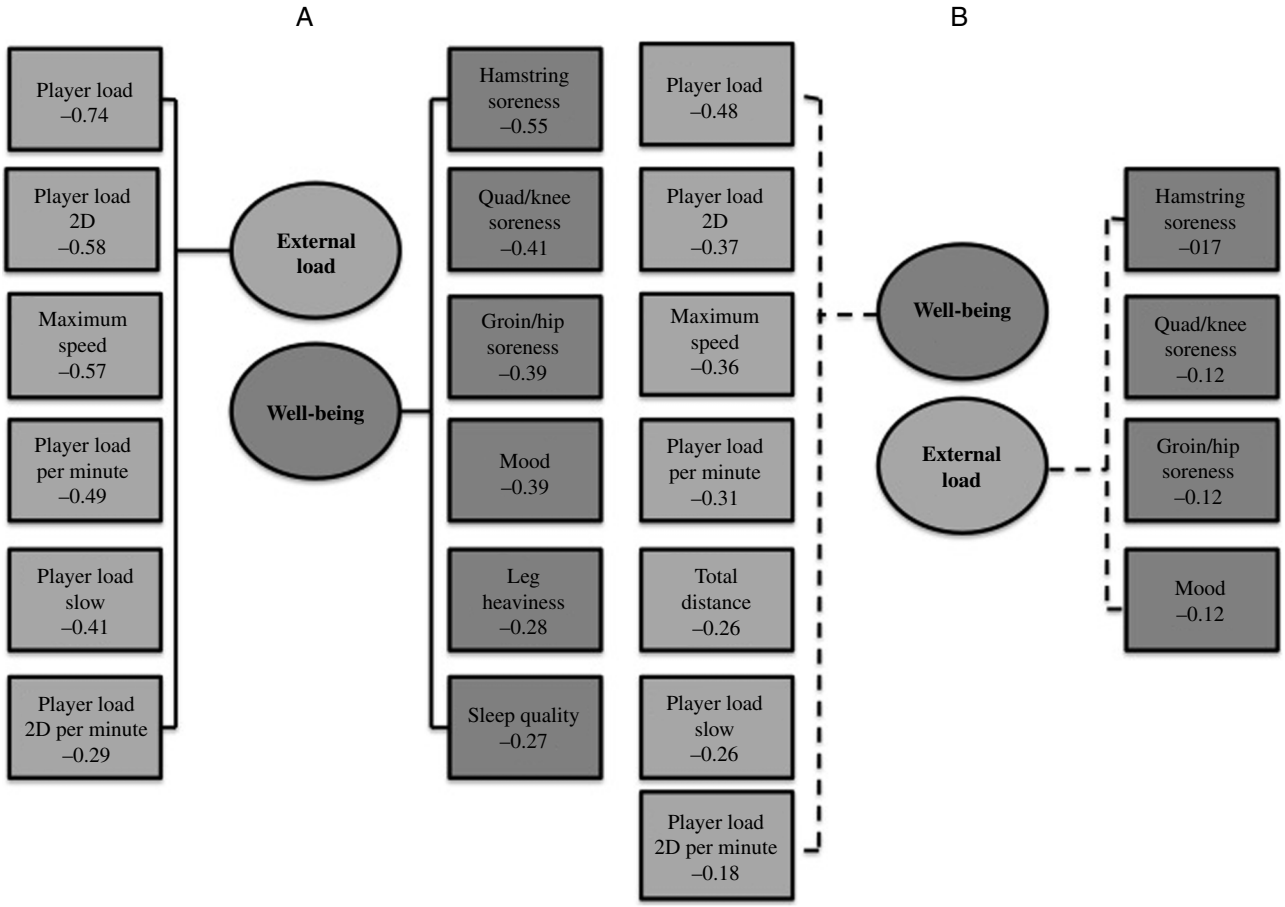

Figure 6 — (A) Well-being and external training load dimension 1* ($r = .67$, $P = 3.6 \times 10^{-58}$) and (B) well-being and external training load dimension 1 cross-loadings. *The dimension is statistically significant ($P < .05$).

Table 5 Well-Being and External Training Load Canonical Correlations

	Dimension 2[a] ($r = .48$, $P = 1.69 \times 10^{-25}$)		Dimension 3[a] ($r = .42$, $P = 3.29 \times 10^{-14}$)		Dimension 4[a] ($r = .37$, $P = 1.21 \times 10^{-6}$)	
	Canonical loading	Cross-loading	Canonical loading	Cross-loading	Canonical loading	Cross-loading
Well-being, AU						
Mood	.63[b]	.13	–.17		–.26	
Calf soreness	.16		–.21		–.12	
Leg heaviness	.47	.10			.12	
Quad/knee soreness	.23		–.29		.25	
Groin/hip soreness	.14		.17		.11	
Lower back soreness	.47	.10			.19	
Sleep duration	.13		–.34[b]		.61[b]	
Energy levels	.51	.11	.24		.18	
Hamstring soreness	.15		–.22			
Sleep quality	.47	.10	–.31		.15	
External load variables						
Absolute loads						
Total distance, m	–.14		.49	.22	–.11	
Player load, AU	–.22	–.11	.37	.16		

(continued)

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC

Table 5 *(continued)*

	Dimension 2[a] ($r = .48$, $P = 1.69 \times 10^{-25}$)		Dimension 3[a] ($r = .42$, $P = 3.29 \times 10^{-14}$)		Dimension 4[a] ($r = .37$, $P = 1.21 \times 10^{-6}$)	
	Canonical loading	Cross-loading	Canonical loading	Cross-loading	Canonical loading	Cross-loading
Player load 2D, AU	−.23	−.12	.37	.16	−.13	
Player load slow, AU	.14		.50	.22	−.13	
Loads relative to training duration						
Maximum speed, km/h	.31[b]	.15	−.11		−.51[b]	−.20
Average velocity, m/min	.23	.12	.52[b]	.23	.26	.10
Player load/min, AU	.28	.14	.23		.28	.11
Player load 2D/min, AU	.27	.12	.12		.14	

Abbreviation: AU, arbitrary unit. Note: The relationship between well-being measures and external training load metrics. Data are expressed as canonical correlations with the canonical loading and cross-loading of each variable presented for each dimension.
[a]The dimension is statistically significant. [b]The variables from each data set that had the strongest contribution to each dimension.

Practical Applications

Pretraining sleep characteristics impact the movement strategy of AF players during main field sessions. Practitioners can expect PL and PL2D to be modified in AF players who have slept poorly. Following training sessions that have high maximum speed, practitioners should maximize the opportunity for sleep following the session by ensuring that subsequent sessions do not start earlier than required. Prioritizing sleep hygiene and recovery strategies to minimize wake episodes may improve the well-being and perceptions of fatigue of AF players. Pretraining well-being may provide some insight into the objective sleep characteristics that may be exhibited in the posttraining sleep opportunity.

Conclusions

Maximum speed, PL, and PL2D were the common associations between both objective sleep characteristics and well-being measures in AF players. This relationship was observed in pretraining and posttraining sleep and well-being measures. Improving pretraining sleep quality and quantity, ideally through minimizing wake time, may have a positive impact on AF players' well-being, movement strategy, and perceptions of fatigue during training. Reviewing maximum speed, total distance, PL, and PL2D from AF main field sessions may provide the ability to modify the schedule (to allow later wake-up time) to assist players need for increased sleep following the session.

References

1. Boyd LJ, Ball K, Aughey RJ. Quantifying external load in Australian football matches and training using accelerometers. *Int J Sports Physiol Perform*. 2013;8(1):44–51. PubMed ID: 22869637 doi:10.1123/ijspp.8.1.44
2. Mooney GM, Cormack JS, O'brien MB, Morgan MW, McGuigan MM. Impact of neuromuscular fatigue on match exercise intensity and performance in elite Australian football. *J Strength Cond Res*. 2013;27:166–173. PubMed ID: 22395264 doi:10.1519/JSC.0b013e3182514683
3. Gallo FT, Cormack JS, Gabbett JT, Lorenzen HC. Self-reported wellness profiles of professional Australian football players during the competition phase of the season. *J Strength Cond Res*. 2017;31(2):495–502. PubMed ID: 27243912
4. Gallo TF, Cormack SJ, Gabbett TJ, Lorenzen CH. Pre-training perceived wellness impacts training output in Australian football players. *J Sports Sci*. 2016;34:1445–1451. PubMed ID: 26637525 doi:10.1080/02640414.2015.1119295
5. Lalor BJ, Halson SL, Tran J, Kemp JG, Cormack SJ. Competition sleep is not compromised compared to habitual in elite Australian footballers. *Int J Sports Physiol Perform*. 2017;13(1):29–36.
6. Juliff LE, Halson SL, Peiffer JJ. Understanding sleep disturbance in athletes prior to important competitions. *J Sci Med Sport*. 2015;18:13–18. PubMed ID: 24629327 doi:10.1016/j.jsams.2014.02.007
7. Pitchford NW, Robertson SJ, Sargent C, Cordy J, Bishop DJ, Bartlett JD. A change in training environment alters sleep quality but not quantity in elite Australian rules football players. *Int J Sports Physiol Perform*. 2016;12(1):75–80. doi:10.1123/ijspp.2016-0009
8. Thornton HR, Delaney JA, Duthie GM, Dascombe BJ. Effects of preseason training on the sleep characteristics of professional rugby league players. *Int J Sports Physiol Perform*. 2018;13(2):176–182. PubMed ID: 28530487 doi:10.1123/ijspp.2017-0119
9. Sargent C, Lastella M, Halson SL, Roach GD. The validity of activity monitors for measuring sleep in elite athletes. *J Sci Med Sport*. 2016;19(10):848–853. PubMed ID: 26794719 doi:10.1016/j.jsams.2015.12.007
10. Sadeh A. The role and validity of actigraphy in sleep medicine: an update. *Sleep Med Rev*. 2011;15:259–267. PubMed ID: 21237680 doi:10.1016/j.smrv.2010.10.001
11. Scott TUM, Scott JT, Kelly GV. The validity and reliability of global positioning systems in team sport: a brief review. *J Strength Cond Res*. 2016;30:1470–1490. PubMed ID: 26439776 doi:10.1519/JSC.0000000000001221
12. Saw AE, Gastin PB, Main LC. Monitoring athletes through self-report: factors influencing implementation. *J Sports Sci Med*. 2014;14:137–146.
13. Sherry A, Henson RK. Conducting and interpreting canonical correlation analysis in personality research: a user-friendly primer. *J Pers Assess*. 2005;84:37–48. PubMed ID: 15639766 doi:10.1207/s15327752jpa8401_09
14. Cohen J. *Statistical Power Analysis for the Behavioural Sciences*. Hillsdale, NJ: Lawrence Erlbaum; 1988.
15. Yeatts PE, Barton M, Henson RK, Martin SB. The use of structure coefficients to address multicollinearity in sport and exercise science.

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC

Meas Phys Educ Exerc Sci. 2017;21:83–91. doi:10.1080/1091367X.2016.1259162
16. R Core Team. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing; 2018.
17. Hadley Wickham RF, Henry L, Müller K. *dplyr: A Grammar of Data Manipulation: R Package Version 0.7.6*. 2018.
18. Wickham H. Reshaping data with the reshape Package. *J Stat Softw*. 2007;21:1–20. doi:10.18637/jss.v021.i12
19. Korkmaz S, Goksuluk D, Zararsiz G. MVN: an R package for assessing multivariate normality. *R J*. 2014;6:151–162. doi:10.32614/RJ-2014-031
20. Gonzalez I, Dejean S, Martin PGP, Baccini A. CCA: Canonical Correlation Analysis: R Package Version 1.2. *J Stat Softw*. 2008;23(12)
21. Porkka-Heiskanen T, Zitting KM, Wigren HK. Sleep, its regulation and possible mechanisms of sleep disturbances. 2013; 208(4):311–328.
22. Wilson MR, Walsh V, Parkin B. *Progress in Brain Research*. Amsterdam, The Netherlands: Academic Press; 2017.
23. Mah CD, Mah KE, Kezirian EJ, Dement WC. The effects of sleep extension on the athletic performance of collegiate basketball players. *Sleep*. 2011;34:943–950. PubMed ID: 21731144 doi:10.5665/SLEEP.1132
24. Killer SC, Svendsen IS, Jeukendrup AE, Gleeson M. Evidence of disturbed sleep and mood state in well-trained athletes during short-term intensified training with and without a high carbohydrate nutritional intervention. *J Sports Sci*. 2017;35(14):1402–1410. PubMed ID: 26406911 doi:10.1080/02640414.2015.1085589
25. Fullagar H, Skorski S, Duffield R, Meyer T. The effect of an acute sleep hygiene strategy following a late-night soccer match on recovery of players. *Chronobiol Int*. 2016;33(5):490–505. PubMed ID: 27031035 doi:10.3109/07420528.2016.1149190
26. Dupuy O, Douzi W, Theurot D, Bosquet L, Dugue B. An evidence-based approach for choosing post-exercise recovery techniques to reduce markers of muscle damage, soreness, fatigue, and inflammation: a systematic review with meta-analysis. *Front Physiol*. 2018;9:403. PubMed ID: 29755363 doi:10.3389/fphys.2018.00403
27. Meeusen R, Duclos M, Foster C, et al. *Prevention, Diagnosis and Treatment of the Overtraining Syndrome: Joint Consensus Statement of the European College of Sport Science (ECSS) and the American College of Sports Medicine (ACSM)*. *Eur J Sport Sci*; 2013;13(1), 1-24.

Authenticated t_lambrinides@yahoo.com | Downloaded 03/18/20 12:37 PM UTC